

Experimental and Theoretical Investigation of the Polarization Dependent System Performance in Multimode Optical Fiber Communication Systems

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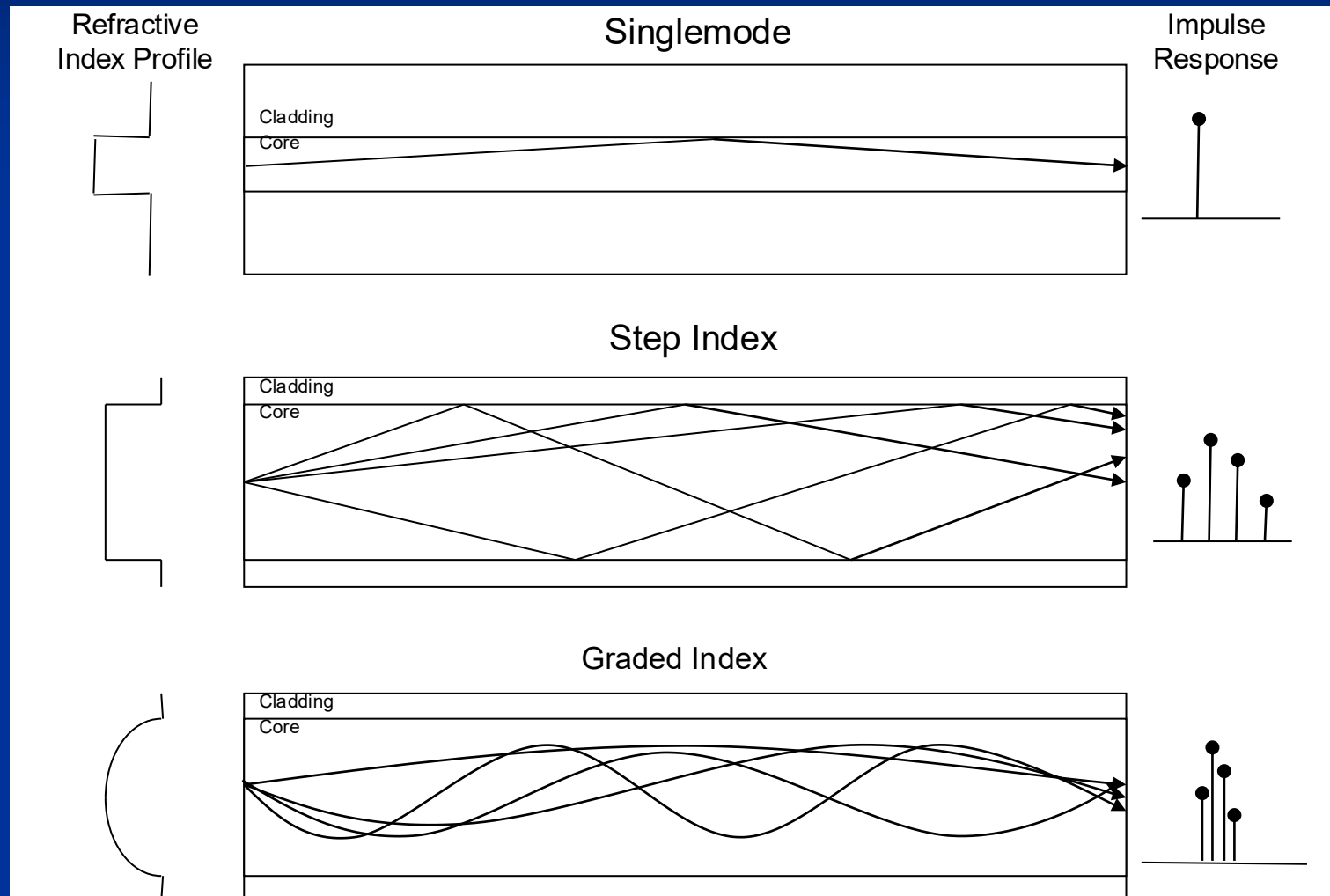
07/05/10



Outline

- Introduction
 - SMF/MMF Comparison
 - Why Choose MMF and Fiber Types
- MMF Impairments
 - Intermodal Dispersion, MSL, Modal Noise
- Polarization dependence
 - Previous Work
 - Experimental Setup/Results
 - Theoretical Setup/Results
- Conclusion

SMF/MMF Comparison



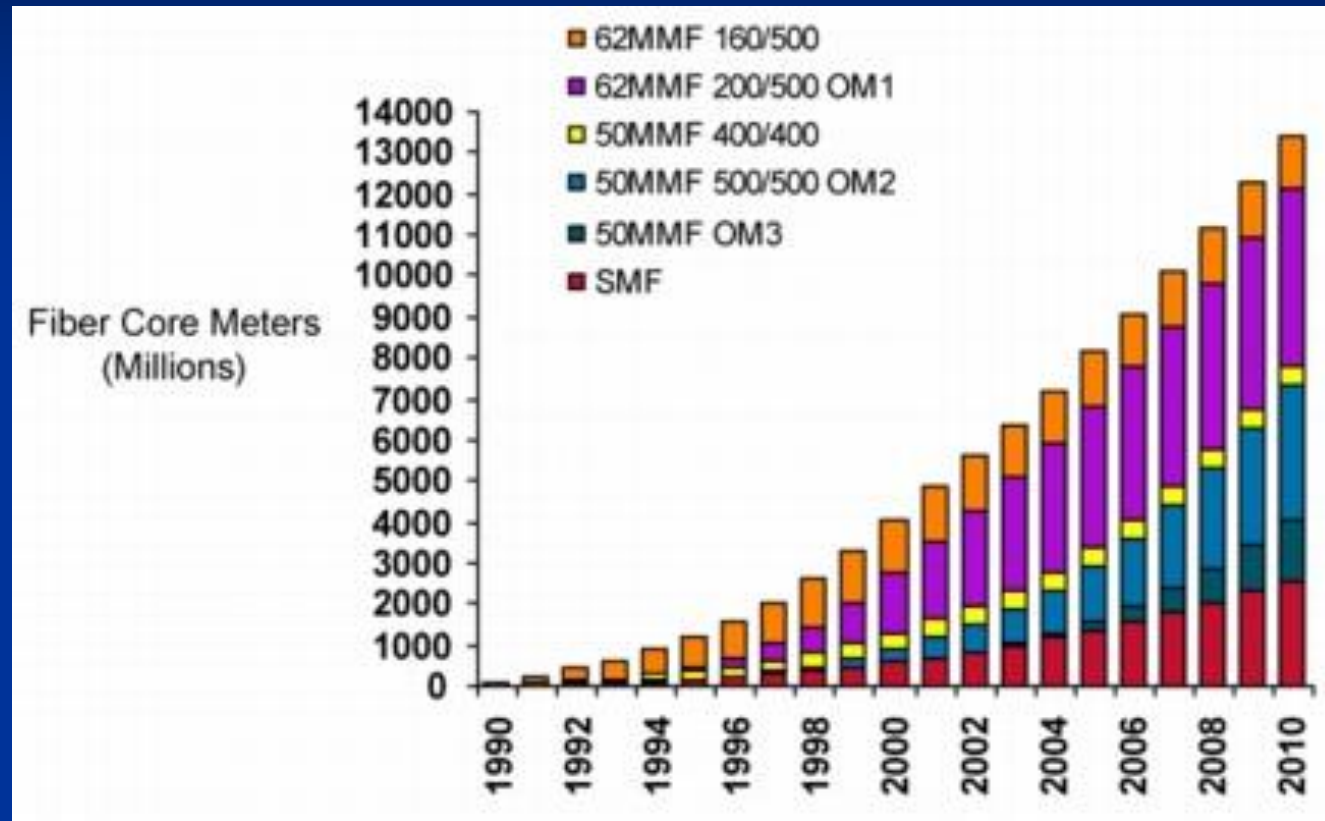
■ Why use Multimode Fiber?

- Large core diameter allows for easy coupling of light into fiber core.

- 50 or 62.5 μm core compared to $\sim 10 \mu\text{m}$ for SMF.
- Relaxes alignment constraints at Tx/Rx and for connectors.
- Lower equipment/labor cost.
 - Can use low cost LEDs or VCSELs.

■ Large installed base for Campus/Enterprise backbone networks.

MMF Installed Base



Azgomi, Houfar. "10GBASE-LRM and EDC: Enabling 10GB Deployment in the Enterprise." White Paper. [Online]. Available: http://www.cisco.com/en/US/prod/collateral/modules/ps5455/prod_white_paper0900aecd806b8bcb.pdf

Flatman, Alan. "In Premises Optical Fiber Installed Base Analysis to 2010." April 2007.

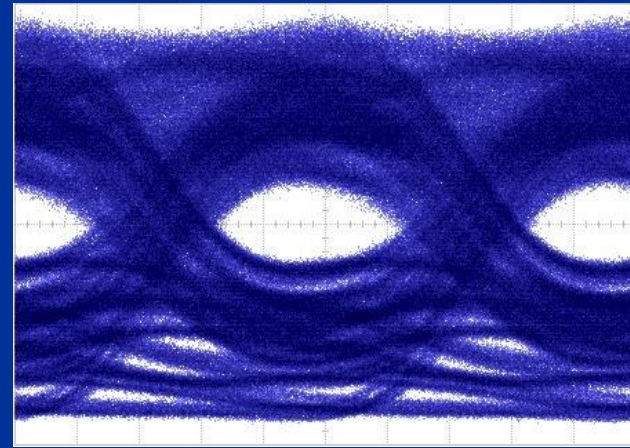
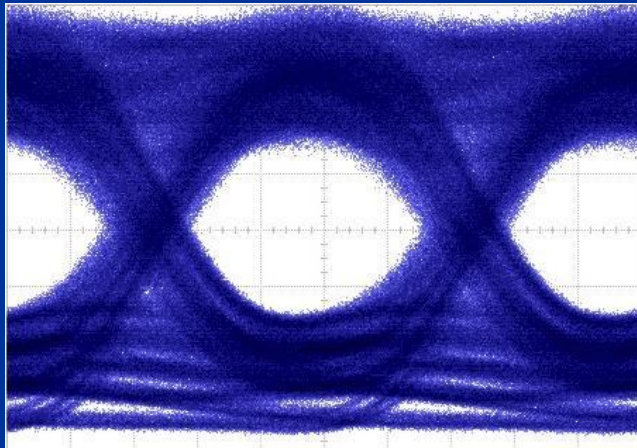
Multimode Fiber Types

Fiber Type	Core Diameter (μm)	OFL BW (MHz.km)
FDDI	62.5	160/500
OM-1	62.5	200/500
OM-2	50	500/500
OM-2+	50	700/500
OM-3	50	1500/500 (EMBW)
OM-3+	50	3500/500 (EMBW)

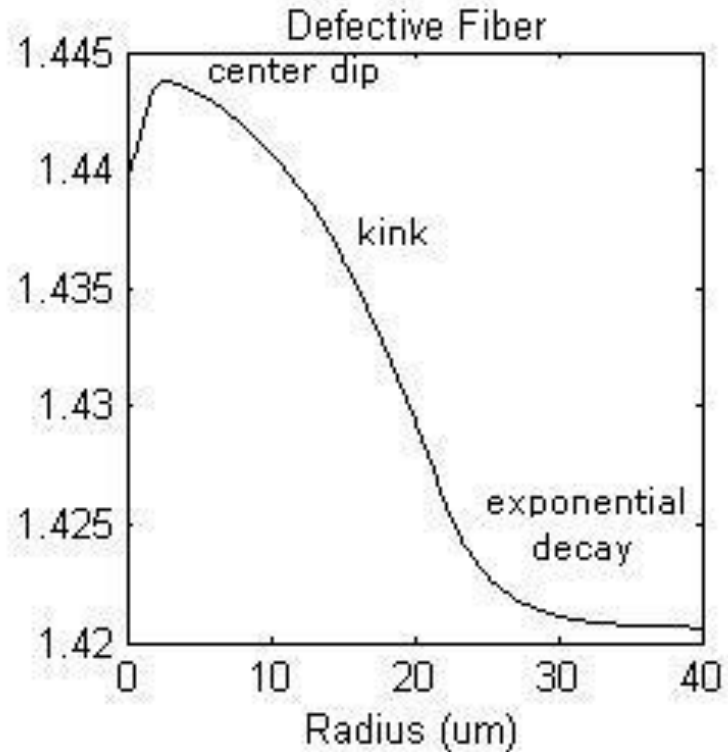
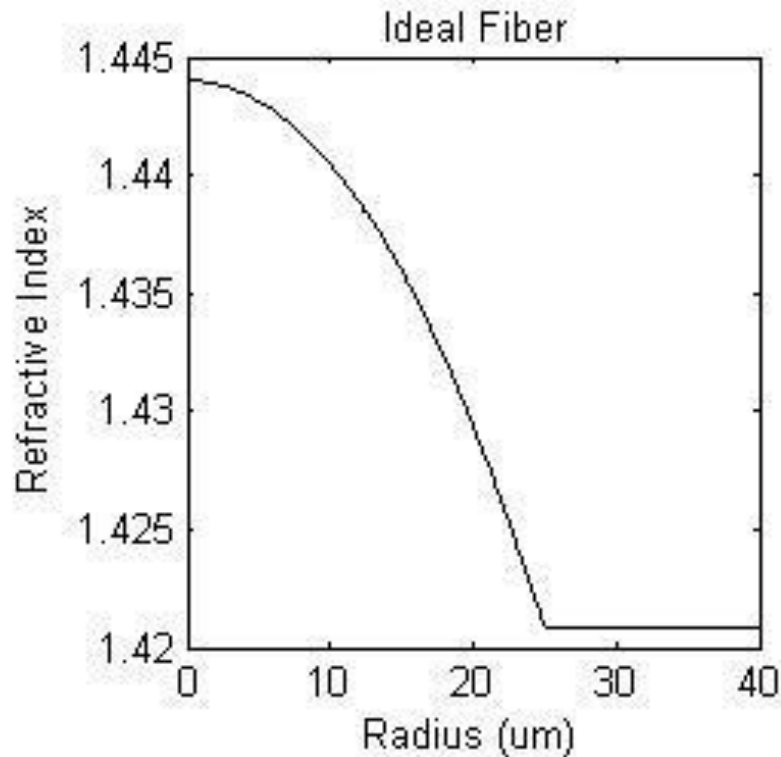
For 850 / 1300 nm sources.

MMF Impairments

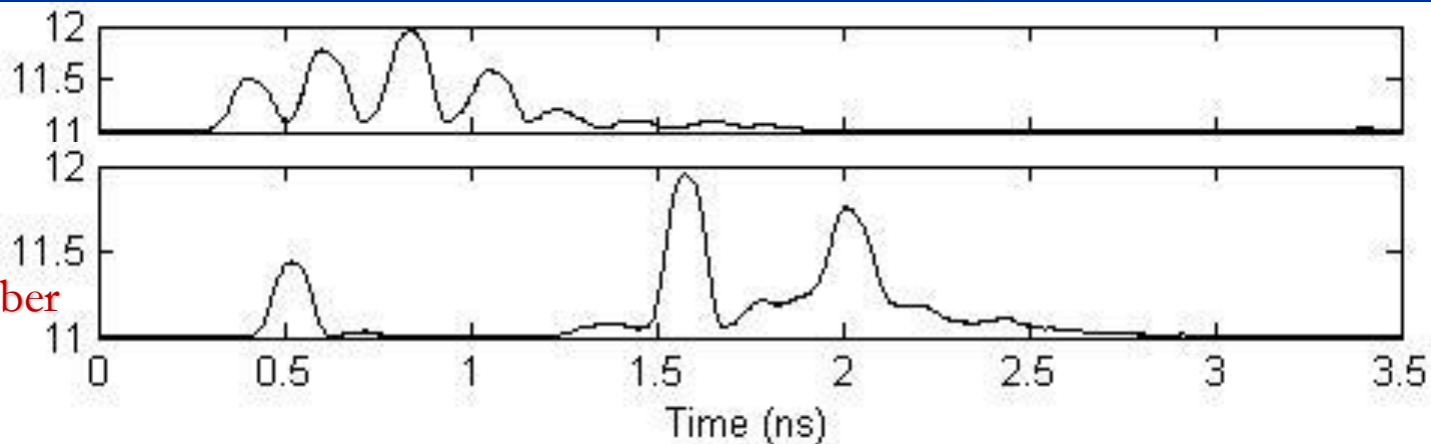
- Refractive Index Defects
- Intermodal Dispersion
- Mode Selective Loss
- Modal Noise
- Polarization Dependence



Fiber Defects

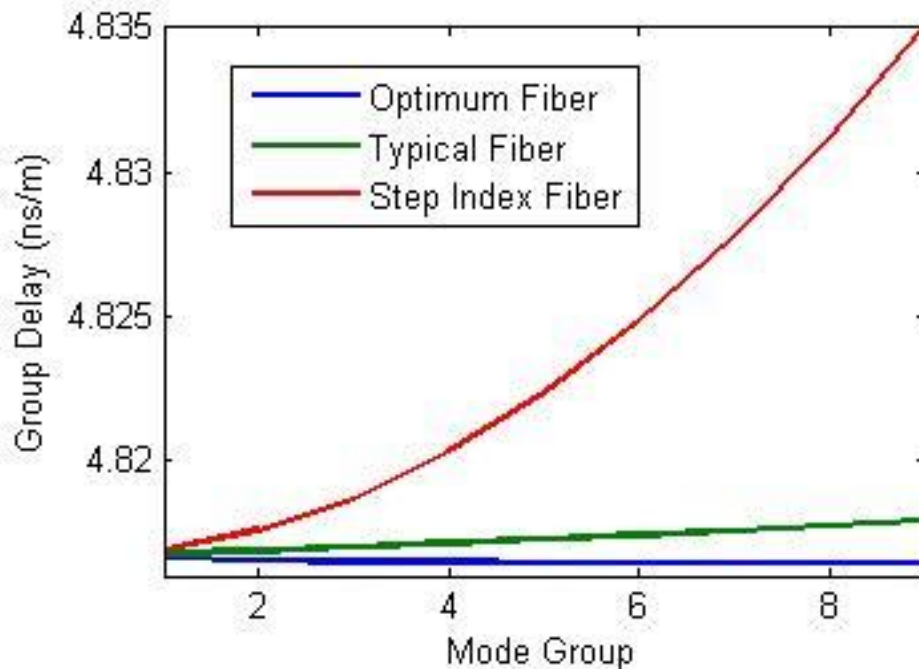
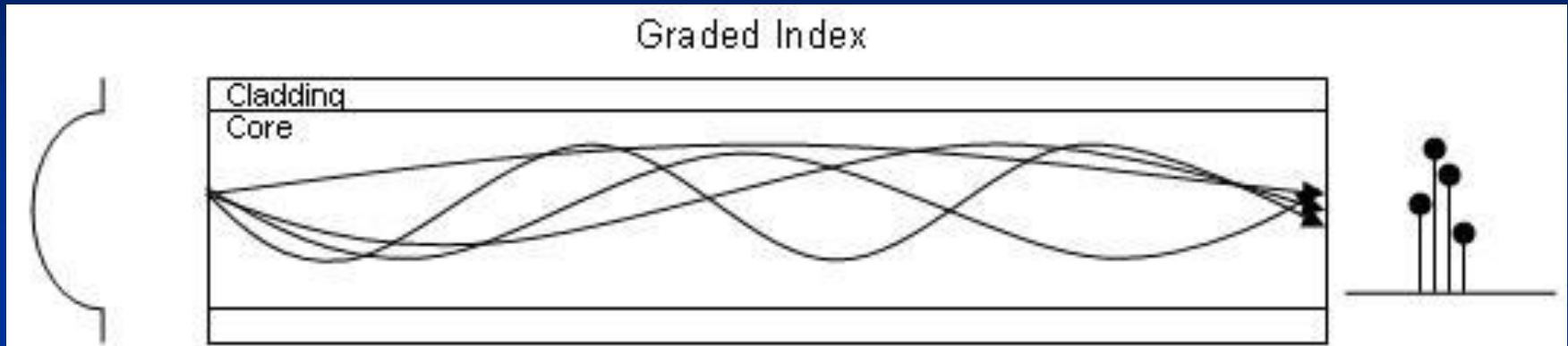


Ideal Fiber



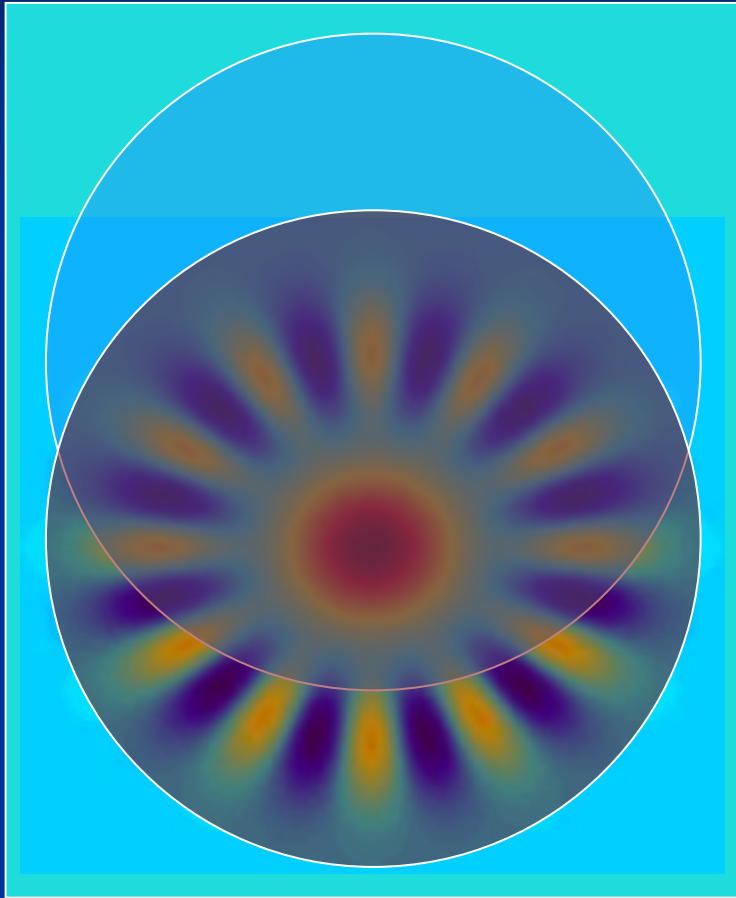
Defective Fiber

Intermodal Dispersion



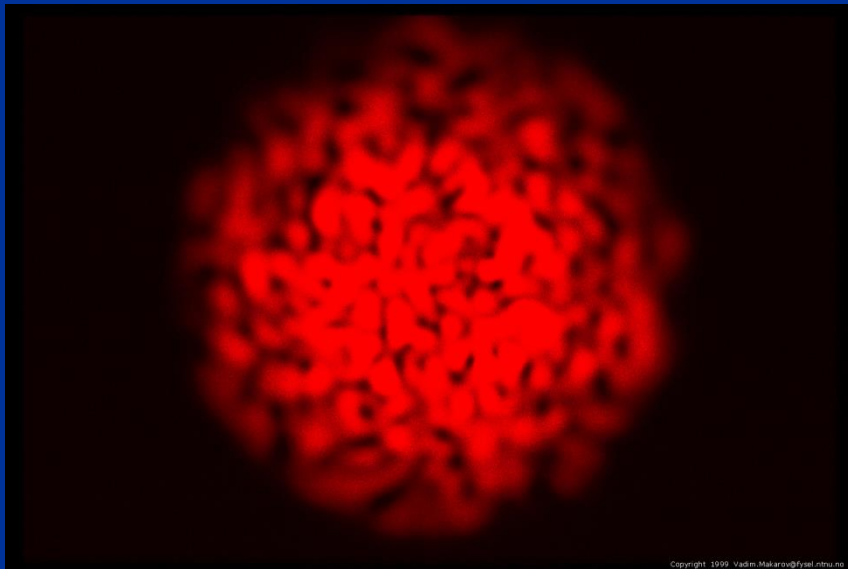
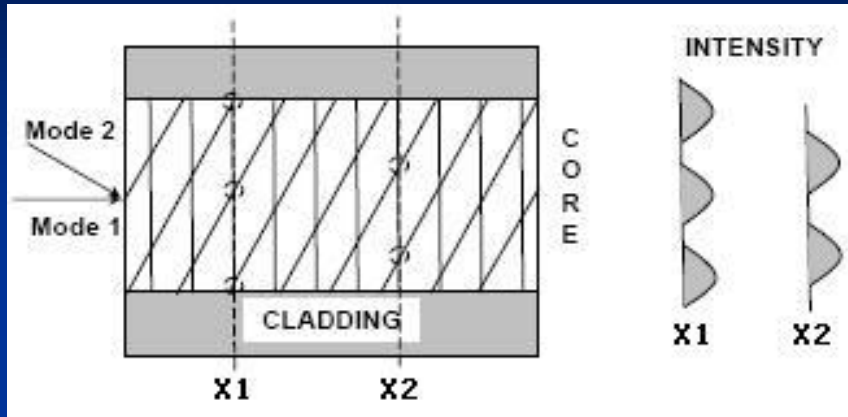
Fiber Type	Maximum Dispersion (ns/km)
step	82.0
parabolic	0.70
optimum	0.17
typical	1.21

Mode Selective Loss



- Mode Selective Loss (MSL)
 - Caused by offset connectors, bends in fiber.
- Leads to Modal Noise

Modal Noise

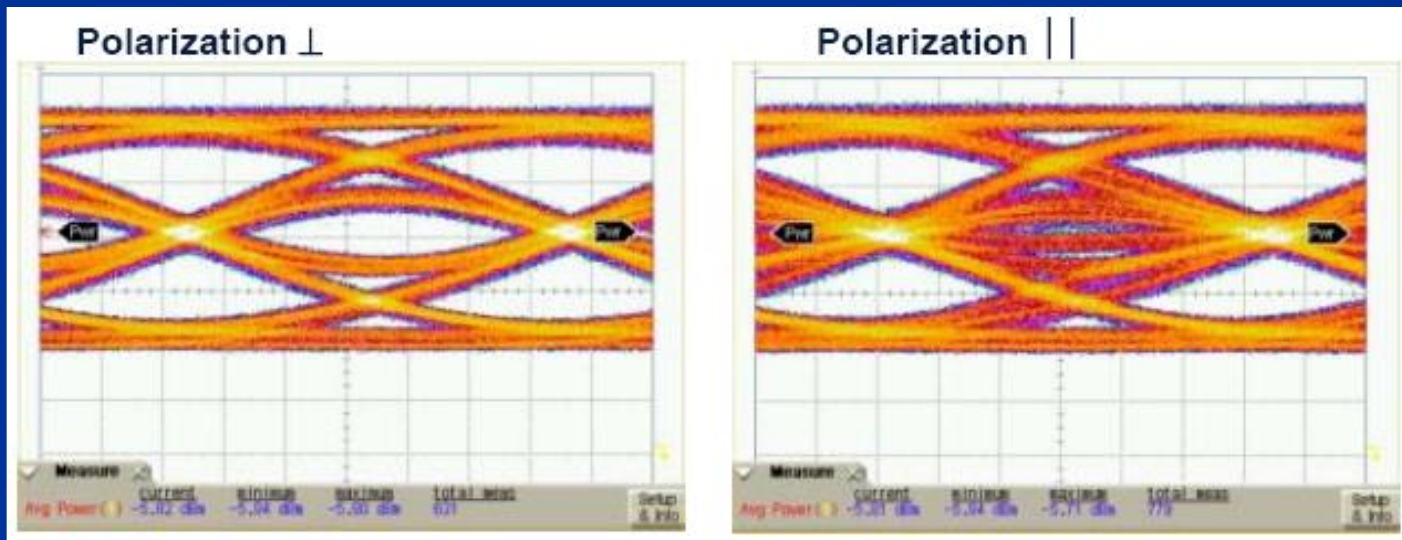


- Coherent source
 - High contrast speckle pattern.
- Dynamic speckle pattern
 - Caused by fiber vibration or heating.
- A source of Mode Selective Loss then causes Modal Noise

Polarization Dependence

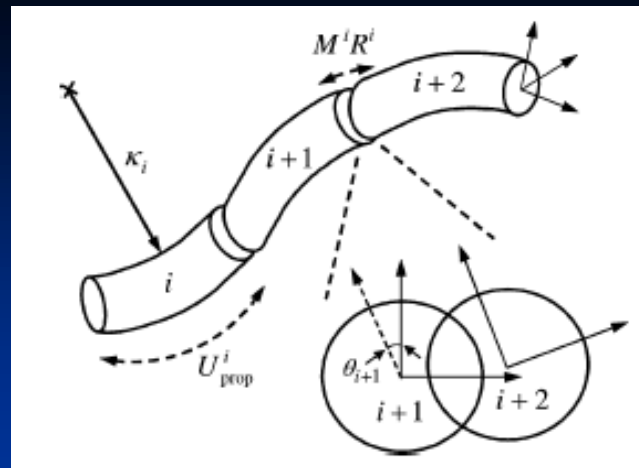
Previous Work

Launch → Fiber ↓	Controlled Offset (17-23μm) (10-16μm)	Offset patchcord (17-23μm) (10-16μm)
Siecor 62.5μm	STRONG	STRONG
Corning OM3	WEAK ↔ STRONG	STRONG



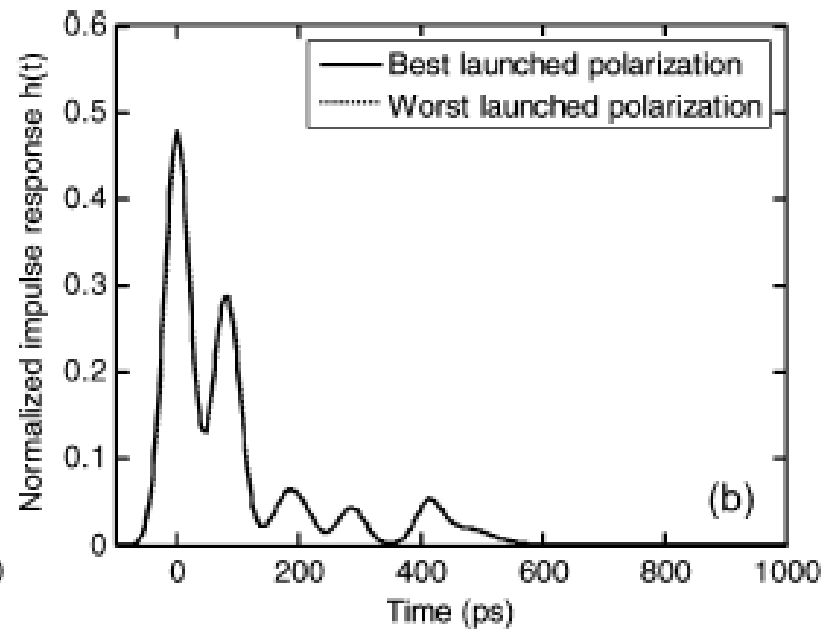
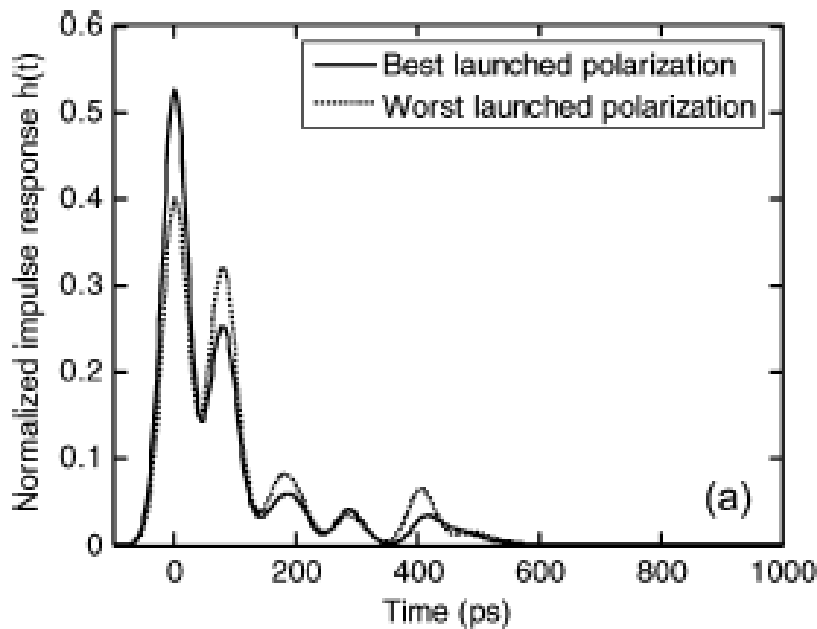
J. Kropp et al. "Polarization Effects in Multimode Fiber Transmission." Presented at the IEEE 802.3aq 10GBASE-LRM Task Force meeting, July 2004. [Online]. Available: http://www.ieee802.org/3/aq/public/upload/kropp_1_1004.pdf

Previous Work



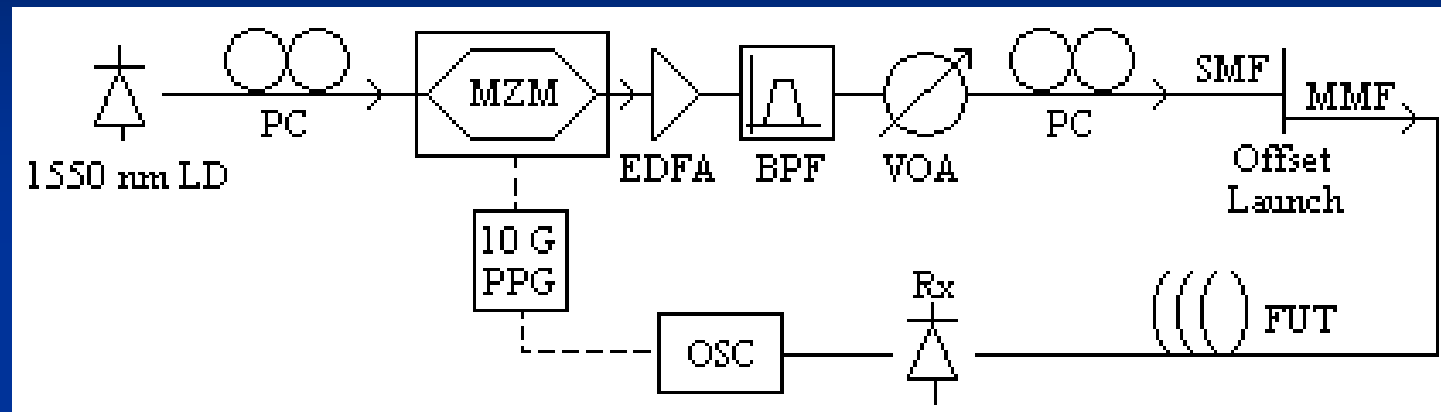
Random Fiber Section Parameters

Identical Section Parameters



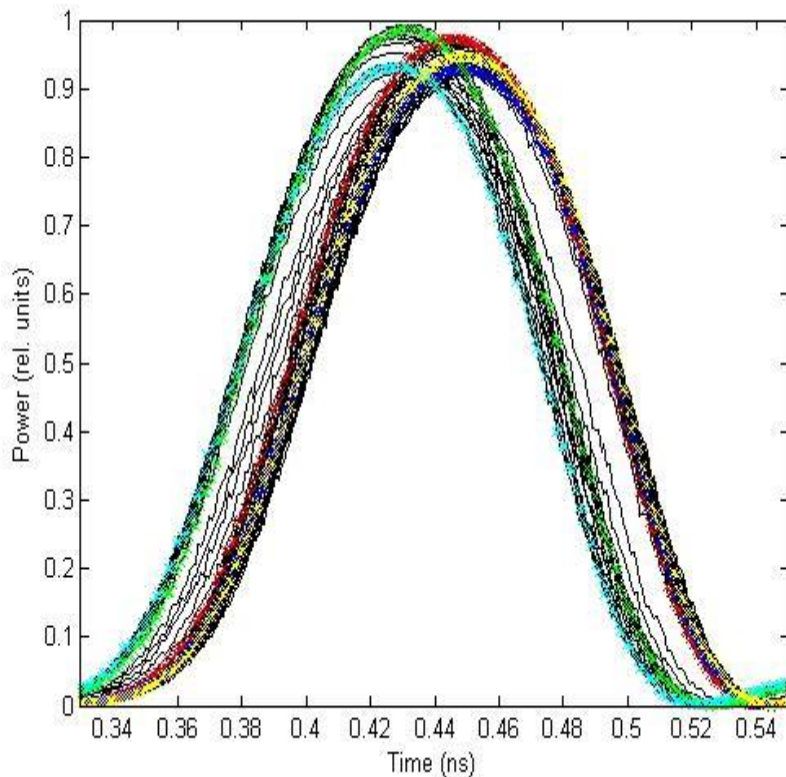
M. Shemirani et al. "Principle Modes in Graded-Index Multimode Fiber in Presence of Spatial- and Polarization-Mode Coupling." J. Light. Tech. vol. 27, No. 10, pp. 1248 – 1261. May 2009.

Experimental Setup

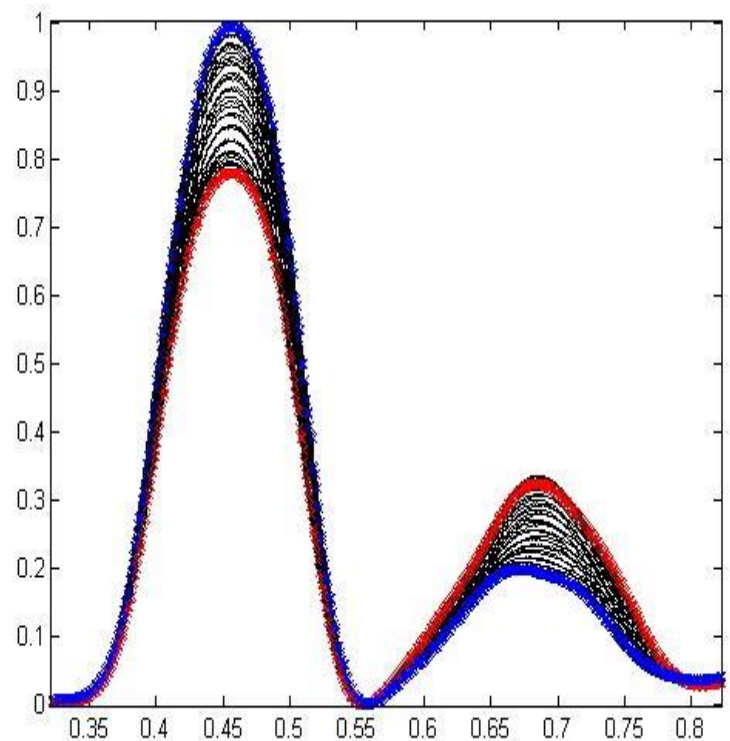


- Fiber Under Test (FUT) can consist of:
 - Single fiber spool, either FDDI or OM3 fiber
 - 1 m long patchcord connected to OM3 fiber
- FUT lengths were 750 m – 1 km
 - Allowed resolution of individual modes/groups at 10 Gb/s

Experimental Results



FDDI

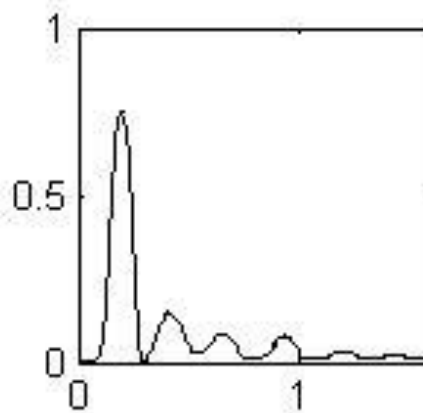
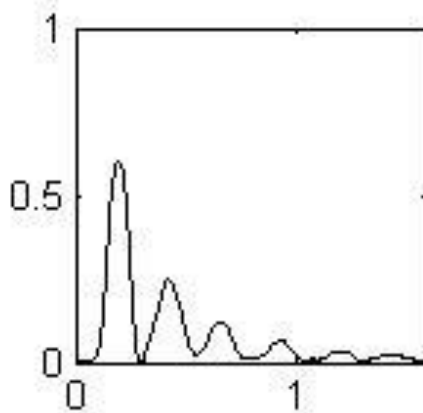
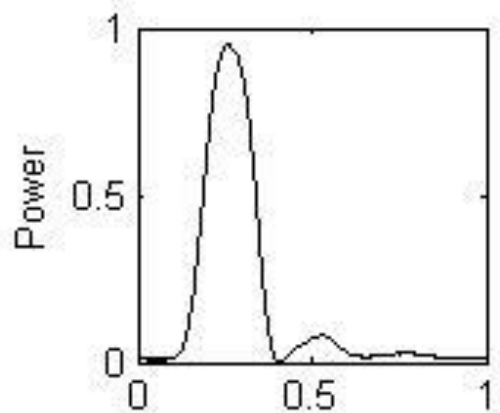
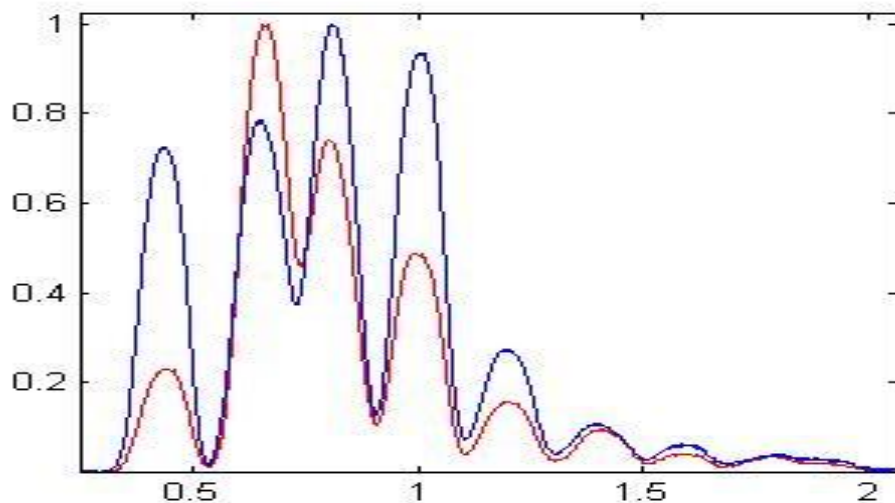


1 m Patchcord - OM3

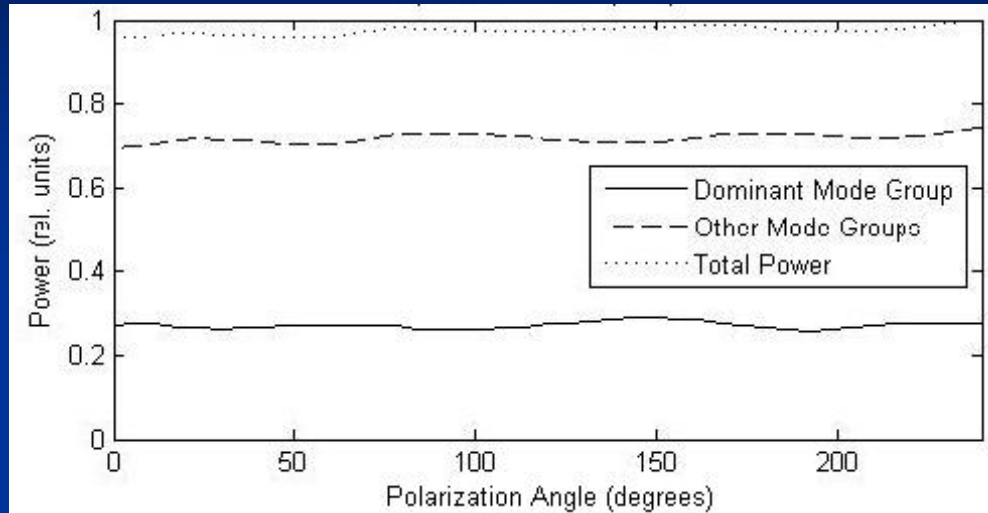
Polarization Rotated from 0 – 240°

Experimental Results

- 1 m Patchcord – OM3 Fiber Spool
- Red
 - 0° input polarization
- Blue
 - 90° input polarization

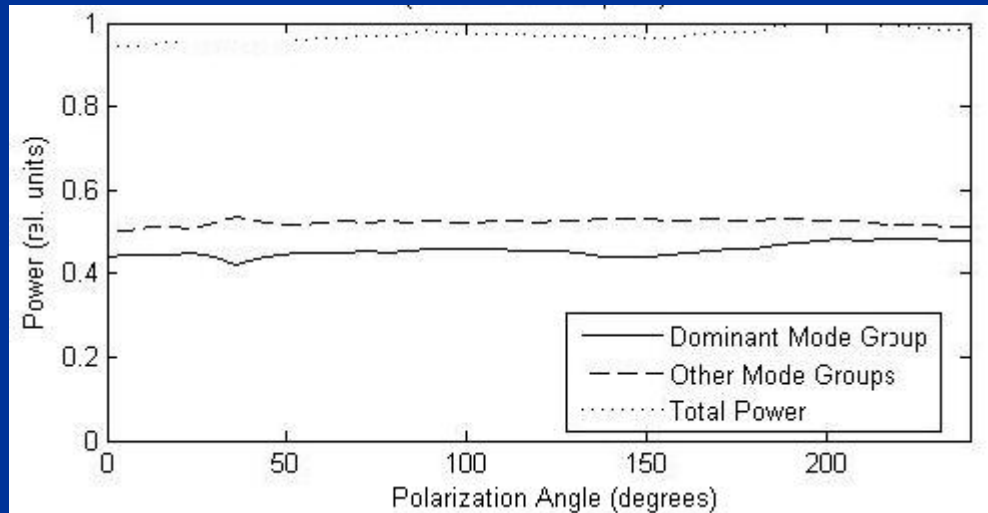


Experimental Results



■ OM3 Fiber Spool

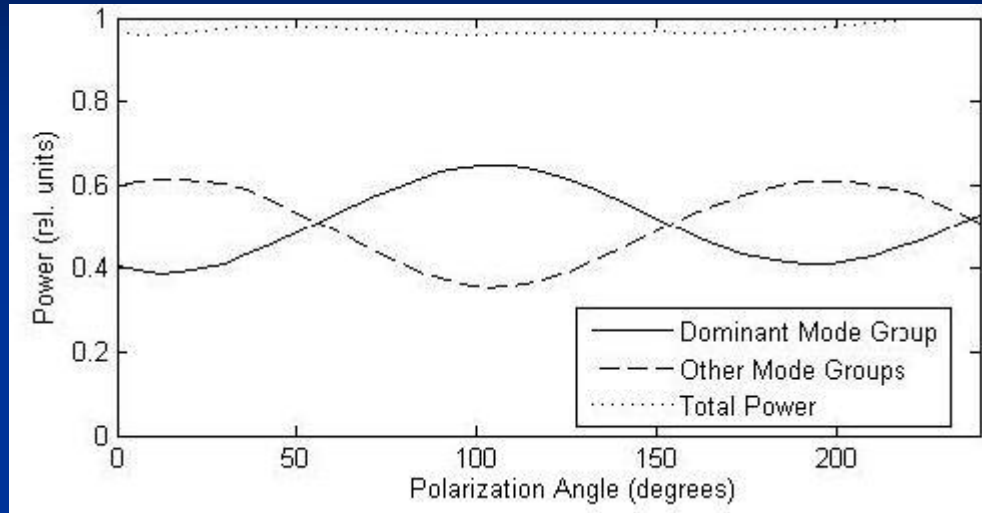
- Very little dependence
- No PDL



■ FDDI Fiber Spool

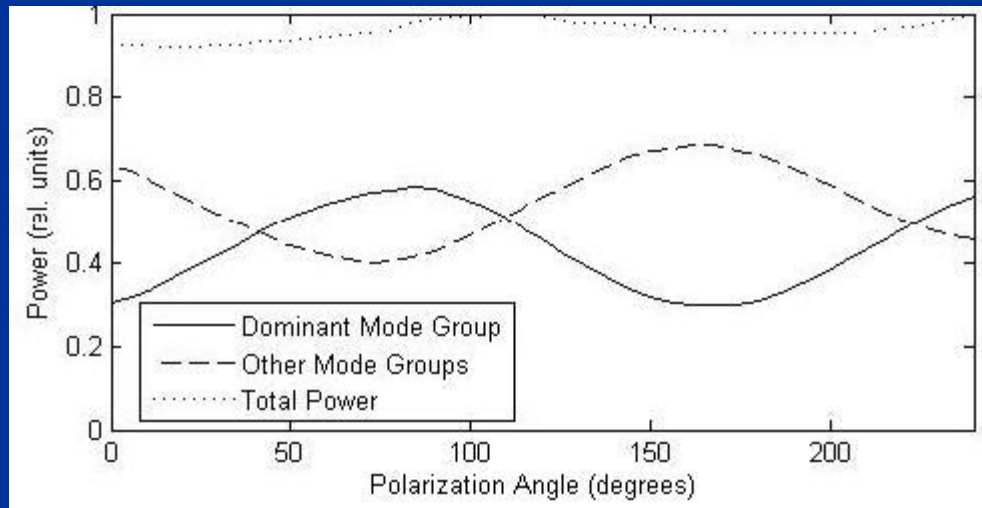
- Very little dependence
- No PDL

Experimental Results



- 1 m Patchcord – OM3+ Fiber Spool

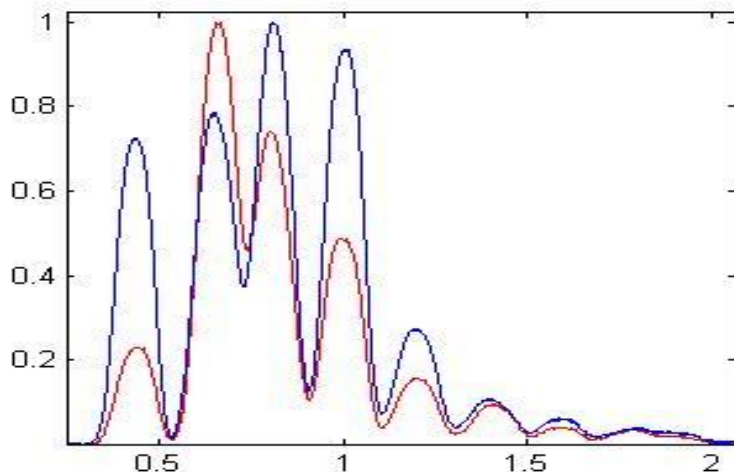
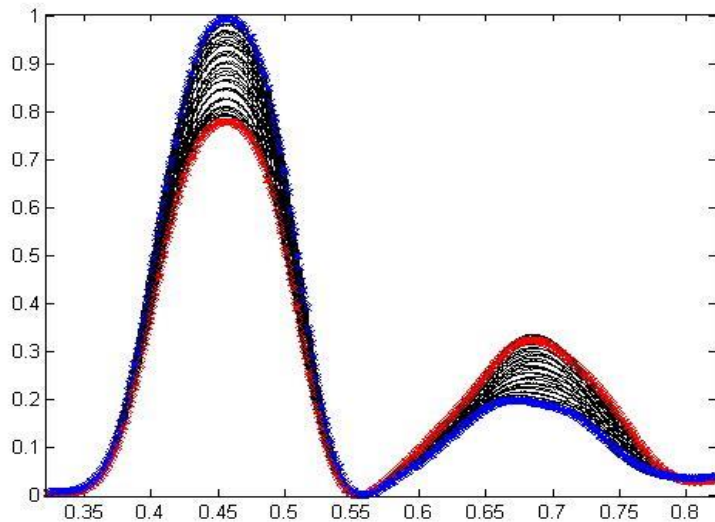
- Very strong dependence
- No PDL



- 1 m Patchcord – OM3 Fiber Spool

- Very strong dependence
- No PDL

Conclusions

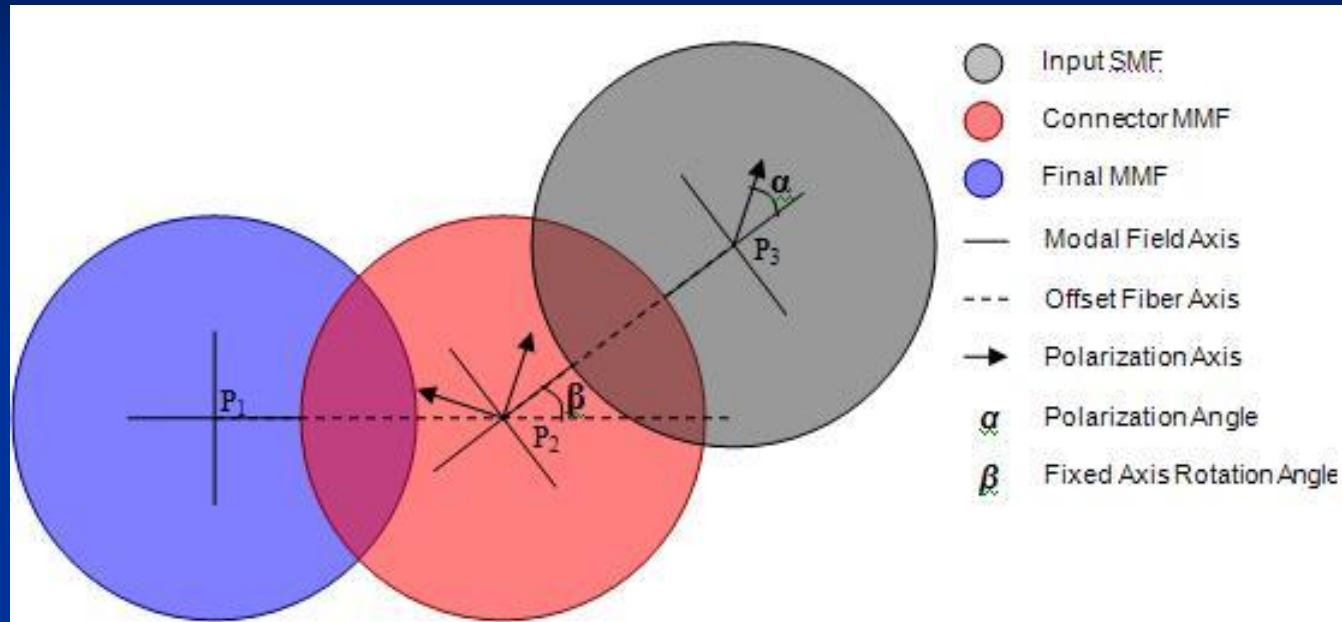


- No polarization dependent loss:
 - No MSL
- No Birefringence:
 - polarization mode coupling
- Significant coupling between mode groups
- No pulse spreading
 - Coupling occurs at beginning of fiber

Possible Causes

- Connector offset
 - Offset patchcord causes mode coupling, maybe polarization dependent
- Patchcord parameter mismatch
 - Parameter mismatch causes mode coupling, maybe polarization dependnent
- Fusion Splice
 - Perturbation caused by a fusion splice might break mode degeneracy causing mode coupling

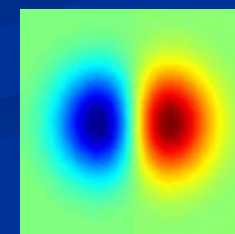
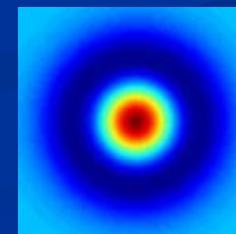
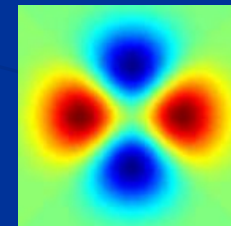
Theoretical Setup



$$r = \sqrt{x^2 + y^2}, \quad \theta = \text{atan} 2\left(\frac{y}{x}\right);$$

$$r' = \sqrt{(x - x_2)^2 + (y - y_2)^2}, \quad \theta' = \text{atan} 2\left(\frac{y}{x - x_2}\right) - \beta;$$

$$r'' = \sqrt{(x - x_3)^2 + (y - y_3)^2}, \quad \theta'' = \text{atan} 2\left(\frac{y - y_3}{x - x_3}\right) - \beta.$$



Ideal Fiber Excitation (SM – MM)

$$\mathbf{R}(\alpha) = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

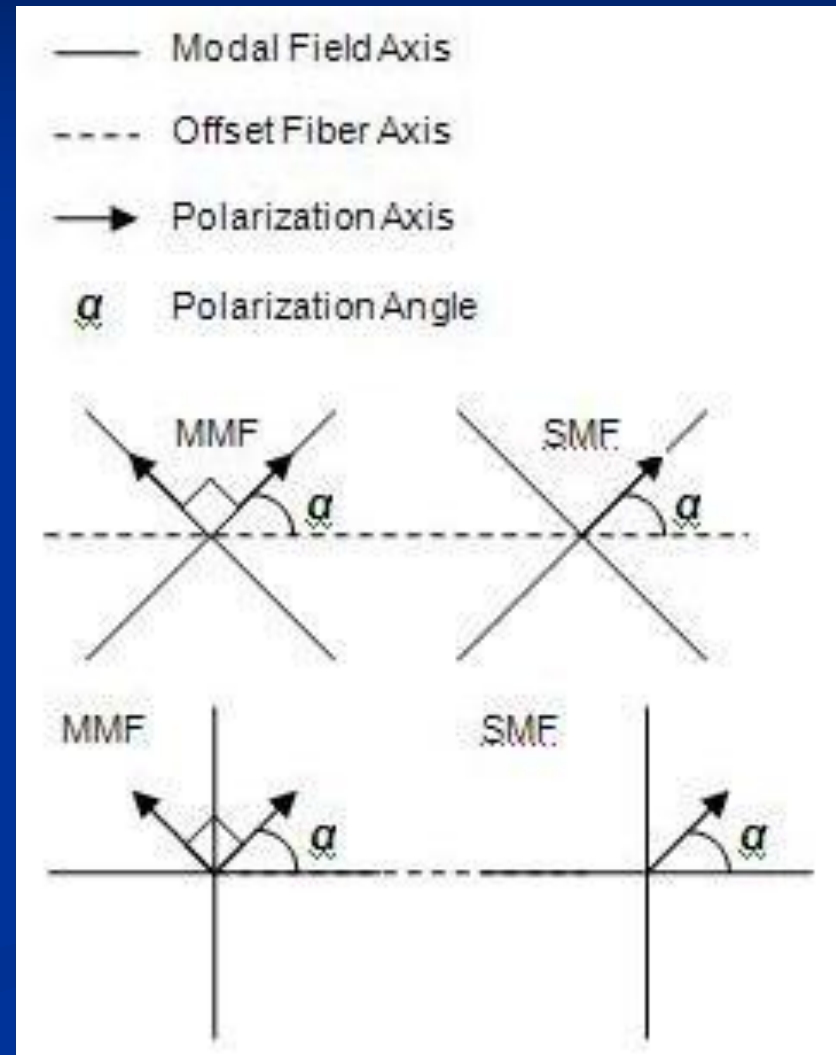
$$k^{lm} = \iint_{r\theta} \psi^{lm} \cdot \phi r dr d\theta.$$

$$\psi^{lm} = \begin{bmatrix} \cos(l\theta - \alpha) \\ \cos(l(\theta - \pi/2) - \alpha) \end{bmatrix} \psi^{lm}(r), \quad \phi = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \phi(r')$$

$$\mathbf{R}(\alpha) \psi^{lm} \cdot \mathbf{R}(\alpha) \phi = \cos(l\theta - \alpha) \psi^{lm}(r) \phi(r')$$

$$\psi^{lm}(r, \theta) = \begin{bmatrix} \cos(l\theta) \\ \cos(l(\theta - \pi/2)) \end{bmatrix} \psi^{lm}(r), \quad \phi = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \phi(r')$$

$$\mathbf{R}(\alpha) \psi^{lm} \cdot \mathbf{R}(\alpha) \phi = \cos(l\theta) \psi^{lm}(r) \phi(r')$$

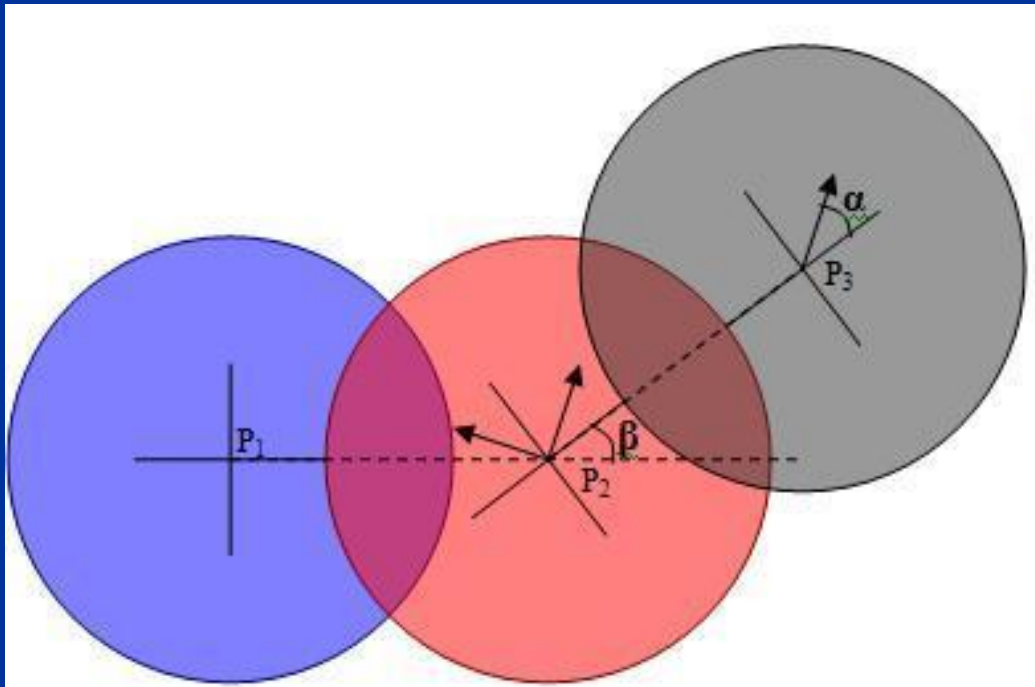


Ideal Fiber Excitation (MM – MM)

$$\frac{da_1}{dz} = -j(\beta_1 + K_{11})a_1 - jK_{12}a_2,$$

$$\frac{da_2}{dz} = -j(\beta_2 + K_{22})a_2 - jK_{21}a_1,$$

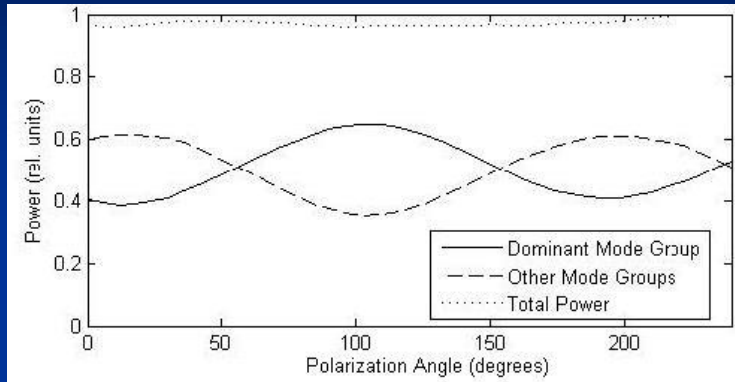
Wei-Ping Huang, “Coupled-mode theory for optical waveguides: an overview.” J. Opt. Soc. Am. A. Vol. 11, No. 3. March 1994. pp 963 - 983



$$\frac{d}{d\alpha} \mathbf{K} = j\mathbf{H}\mathbf{K}$$

$$\mathbf{K} = \begin{bmatrix} k_1 \\ \vdots \\ k_N \end{bmatrix}; \quad \mathbf{H} = \begin{bmatrix} h_{11} & \cdots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{N1} & \cdots & h_{NN} \end{bmatrix}$$

Ideal Fiber Excitation (MM – MM)



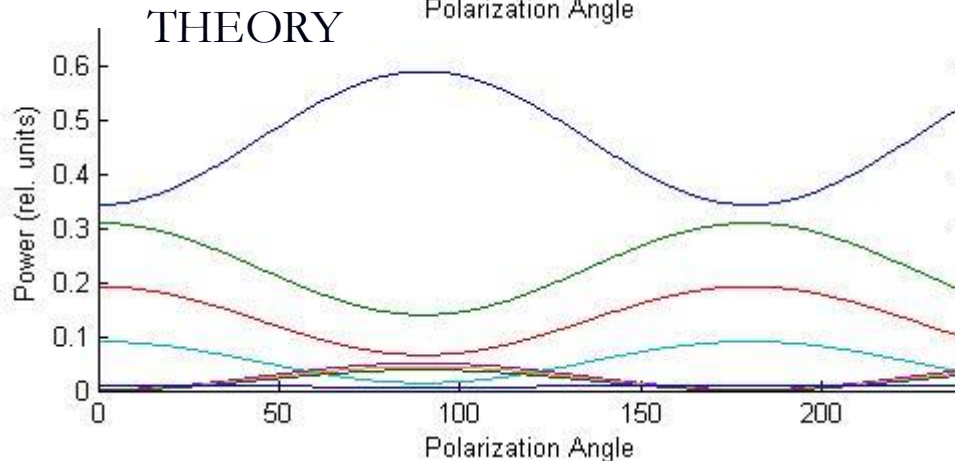
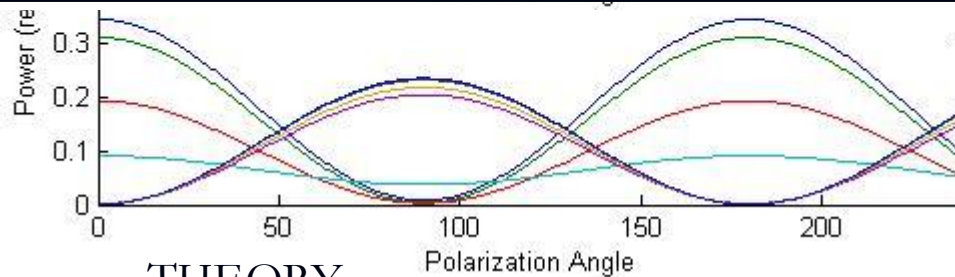
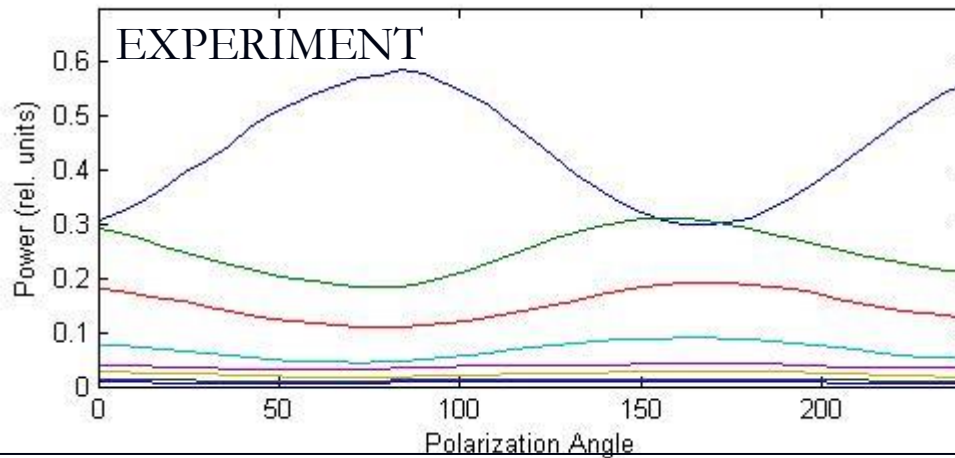
$$P_n(\alpha) = p_{n1} \cos^2(\alpha - \sigma_n) + p_{n2} \sin^2(\alpha - \sigma_n)$$

$$k_n(\alpha) = a_{n1} \cos(\alpha - \sigma_n) + ja_{n2} \sin(\alpha - \sigma_n)$$

$$\mathbf{H} = \begin{bmatrix} h_d & h & h & \dots & h \\ h & h_d & h & \dots & h \\ h & h & h_d & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & h \\ h & h & \dots & h & h_d \end{bmatrix}$$

$$k_n = \exp \left\{ j \left(\frac{N-2}{2} h - h_d \right) \alpha \right\} \left\{ k_n(0) \cos \left(\frac{N}{2} h \alpha \right) + j \frac{1}{N} \left(2 \sum_{m \neq n} k_m(0) - (N-2) k_n(0) \right) \sin \left(\frac{N}{2} h \alpha \right) \right\}$$

$$k_n = \exp\left\{j\left(\frac{N-2}{2}h - h_d\right)\alpha\right\}\left[k_n(0)\cos\left(\frac{N}{2}h\alpha\right) + j\frac{1}{N}\left(2\sum_{m \neq n} k_m(0) - (N-2)k_n(0)\right)\sin\left(\frac{N}{2}h\alpha\right)\right]$$



$ k(0) ^2$	n	Power
	1	0.3443
	2	0.3108
	3	0.1932
	4	0.0911
	5	0.0019
	6	0.0008
	7	0.0002
	8	0.0001

$k(0)$	n	Amplitude
	1	0.5868
	2	0.5575
	3	0.4396
	4	0.3018
	5	0.0431
	6	0.0284
	7	0.0136
	8	0.0107

$k(0)$	n	Amplitude
	1	0.5868
	2	-0.5575
	3	-0.4396
	4	-0.3018
	5	0.0431
	6	0.0284
	7	0.0136
	8	-0.0107

Conclusion

- Shown polarization effect as a sinusoidal power coupling between modes
- Determined effect happens at a discrete point along the fiber
- Possibly caused by offset fiber, parameter mismatch, fusion splice
- Mathematical model created to describe effect, shows good agreement with theory

Any Questions!?

The experiment was conducted at the National Microelectronics and Photonics Testing Collaboratory in Kingston, ON.

